

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, adherent	Molecules Electroporated	DNA: various plasmids, about 3kB.
Species Used	Human, Hep3b2, hepatocytes; HeLa, epithelial carcinoma		

Before the Pulse

Cell Growth Medium	DMEM, MAB87 /3 + 10% serum (GIBCO/BRL, Sigma)	Growth Phase at Harvest	Not given
		Pre-pulse Incubation	5 min.

Wash Solution Not given

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature	25 °C	Cuvette Gap	0.4 cm
Electroporation Medium*	Cell growth medium	Voltage	0. 220 kV
Cell Density	5 x 10 (6) cells / pulse	Field Strength	0.55 kV/cm
Volume of Cells	400 µl	Capacitor	960 µF
DNA Concentration	75 µg / pulse	Resistor	(Pulse Controller) Ω none
DNA Resuspension Buffer	Sterile deionized water	Time Constant	30 msec
Volume of DNA	5 to10 µl		
After the Pulse			
Outgrowth Medium	Same as growth medium		

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature	37 °C
Length of Incubation	Overnight
Selection Method or Assay Used	b-gal staining (X-gal) luciferase
Electroporation Efficiency	52%
Per Cent Survival	90%

Name of Submitter Margie Wilde, Research Assistant

Institution Address Texas Children's Hospital
Pathology Department
6621 Fannin
Houston, Texas 77030

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